

Project SHIVAM - Notes 01 (HTML & CSS Foundations)

HTML vs CSS

HTML defines the structure of a webpage. CSS controls its appearance.

Important HTML Elements

<!DOCTYPE html> - tells browser this is HTML5.
<html> root element.
<head> page information.
<body> visible content.
<header> top section.
<nav> navigation.
<main> primary content.
<section> logical section.
<div> generic container.
 inline container.
<form>, <input>, <button> for user input.
<a> creates hyperlinks.

Elements vs Attributes

Elements are tags like div, header, nav. Attributes give extra information like class, id, href, src, type, placeholder.

Meta Tags

charset=UTF-8 supports almost all languages. viewport makes the page use the device width on phones and tablets.

DOM

Browser converts HTML into a DOM Tree. JavaScript later works with this DOM instead of raw HTML.

Semantic HTML

Use header, nav, main, section, footer because they describe meaning and improve accessibility and SEO.

CSS Basics

CSS syntax: selector { property: value; }

Selectors

Element selector: body {}
Class selector: .menu {}
Descendant selector: .menu a {}

Important CSS Properties

margin - outside spacing
padding - inside spacing
background/background-color - background
color - text color
font-family - font selection
width - width
border-radius - rounded corners
cursor:pointer - hand cursor
transition - smooth animation

Flexbox

display:flex creates a flex layout.
justify-content controls horizontal distribution.
align-items controls vertical alignment.
gap creates spacing between flex children.
flex:1, flex:2 are ratios for sharing available space.

Current Project Progress

Completed: Header, Logo, Search Bar, Navigation, Login, Orders, Cart, Hero Section.
Next: Professional Header, Hero Banner, Product Cards, CSS Grid, Responsive Design.